

Day 02 - Networking Fundamentals

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Purpose	Understand core networking concepts and identify the network configuration of my own Windows computer.
Practical Commands	ipconfig, ping, and tracert
Environment	Personal Windows computer / local network

1. THEORY

1. What is a computer network? A computer network is a group of connected computers and devices that communicate and share resources such as files, applications, printers, and Internet access.	7. What is a firewall? A firewall is a security control that monitors and filters network traffic according to defined rules. It can allow legitimate traffic and block unauthorized or unwanted traffic.
2. What is the difference between a client and a server? A client is a device or application that requests a service. A server is a system that provides a service or resource to clients. A web browser is a client and a web server provides web pages.	8. What is an IP address? An IP address is a logical address assigned to a network device so it can be identified and communicate over an IP network. IPv4 uses four decimal numbers separated by dots.
3. What is a LAN? LAN stands for Local Area Network. It connects devices within a limited area such as a home, office, school, or building.	9. What is a MAC address? A MAC address is a hardware-level address used to identify a network interface on a local network. It is normally written as six hexadecimal pairs.
4. What is a WAN? WAN stands for Wide Area Network. It connects networks across large geographical areas. The Internet is the best-known example of a WAN.	10. What is a default gateway? A default gateway is usually the router that a computer sends traffic to when the destination is outside the local network.
5. What is the purpose of a switch? A switch connects devices within a local network and forwards Ethernet frames to the correct device using MAC addresses.	11. What is DNS? DNS stands for Domain Name System. It translates human-readable domain names such as example.com into IP addresses used for communication.
6. What is the purpose of a router? A router connects different networks and forwards IP packets between them. A home router commonly connects a local network to the Internet.	12. Why is networking knowledge important for cybersecurity? Networking knowledge helps security professionals understand IP addresses, ports, protocols, DNS, routing, firewalls, and traffic. These skills are important for detecting attacks, investigating incidents, analyzing packets, and protecting systems.

2. YOUR NETWORK INFORMATION

The values below are taken from the user's supplied Windows **ipconfig** screenshot. The active Wi-Fi adapter is used for the private IPv4 and IPv4 default gateway.

Network Item	Value	Explanation
Private IPv4 Address	10.228.143.3	Local IPv4 address shown for the Wi-Fi adapter.
Subnet Mask	255.255.255.0	Defines the local /24 network.
Default Gateway	10.228.143.240	IPv4 gateway shown under the Wi-Fi adapter.
DNS Server	Not visible in supplied screenshot	Run ipconfig /all to display and record the DNS server.

Practical command notes

Command	Purpose
ipconfig	Shows local IP configuration, including IPv4 address, subnet mask, and default gateway.
ping 8.8.8.8	Tests basic IP connectivity to a reachable Internet destination.
tracert 8.8.8.8	Shows the network hops taken between the computer and the destination.

DNS note: The supplied screenshot uses **ipconfig**, which does not show the DNS server. The final submission should record the DNS Server value from **ipconfig /all**.

3. SCREENSHOTS - IP CONFIGURATION

The screenshot below is the original output supplied for this report. It shows the Windows network configuration and the active Wi-Fi IPv4 settings.

Screenshot 1 - ipconfig

```
C:\Users\gaura>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::863d:7967:6981:dbd%3
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:62c1:d3d:151b:a07:9a7:46ea
    Temporary IPv6 Address. . . . . : 2401:4900:62c1:d3d:652b:1924:8668:86c8
    Link-local IPv6 Address . . . . . : fe80::161a:fab4:a219:8c79%8
    IPv4 Address. . . . . : 10.228.143.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::3c3f:7bff:fee9:e1ba%8
                                10.228.143.240

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

Observation: Active Wi-Fi IPv4 = 10.228.143.3; subnet mask = 255.255.255.0; IPv4 default gateway = 10.228.143.240.

4. REQUIRED CONNECTIVITY TESTS

Screenshot 2 - ping

Run this on the same computer and paste the screenshot into the box:

```
ping 8.8.8.8
```

```
C:\Users\gaura>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=45ms TTL=114
Reply from 8.8.8.8: bytes=32 time=48ms TTL=114
Reply from 8.8.8.8: bytes=32 time=42ms TTL=114
Reply from 8.8.8.8: bytes=32 time=55ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 42ms, Maximum = 55ms, Average = 47ms
```

Screenshot 3 - tracert

Run this on the same computer and paste the screenshot into the box:

```
tracert 8.8.8.8
```

```
C:\Users\gaura>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    4 ms     5 ms     2 ms  10.228.143.240
  2   318 ms   199 ms   203 ms  192.168.29.10
  3    37 ms    22 ms    37 ms  192.168.28.137
  4    45 ms    22 ms    23 ms  192.168.31.7
  5    28 ms    31 ms    30 ms  192.168.31.33
  6     *        *        *    Request timed out.
  7    30 ms    52 ms    29 ms  nsg-corporate-129.100.187.122.airtel.in [122.187.100.129]
```

5. LEARNING SUMMARY

This task helped me understand how computers communicate over networks and the role of common networking components. I learned how IP addresses identify devices, how MAC addresses identify network interfaces on local networks, and how switches, routers, firewalls, gateways, and DNS support communication and security.

Command	What I learned
ipconfig	View the local IP configuration of a Windows computer.
ping	Test whether a destination is reachable over IP.
tracert	Trace the sequence of network hops toward a destination.

Submission Checklist

- ✓ Name + Day 02 shown on page 1
- ✓ All 12 theory questions answered
- ✓ Private IPv4 address entered: 10.228.143.3
- ✓ Default gateway entered: 10.228.143.240
- ✓ DNS server added from ipconfig /all
- ✓ ipconfig screenshot included
- ✓ ping screenshot included
- ✓ tracert screenshot included

Completion note: This PDF includes the supplied ipconfig screenshot and all written answers. Because ping and tracert screenshots were not supplied yet, their sections remain ready for insertion. The DNS server is also not visible in the supplied ipconfig output.